

ALGEBRAIC NUMBER THEORY

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Chapter 1

THE RATIONALS

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Prerequisites for this course include Rings and Modules (tensor products, localization, etc.), Field and Galois Theory, and bits of Commutative Algebra and Algebraic Geometry.

1.1 Algebraic Extensions

The basic object of study in algebraic number theory is the field of rational numbers $\mathbb{Q}$. We recall the field inclusions $\mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$. We will be interested in the field of algebraic numbers $\overline{\mathbb{Q}} \subset \mathbb{C}$.

Definition 1.1. We say $k \subseteq L$ is a *field extension* if k and L are both fields. In this case, we say L is an extension of k and write L/k .

Let k be a field, and $k[X]$ the polynomial ring over k . Let $f(X) \in k[X]$ be a polynomial with $\deg(f) = n$, and $\langle f(X) \rangle \subseteq k[X]$ the ideal generated by $f(X)$. Now define $V := k[X]/\langle f(X) \rangle = \{g(X) + \langle f(X) \rangle \mid g(X) \in k[X]\}$. This V is a vector space over k . Moreover, $\dim_k(V) = n = \deg(f)$; let $\overline{g(X)} = g(\overline{X}) = g(X) + \langle f(X) \rangle$ denote an element in V . Using the division algorithm, we can write $g = fh + r$ with $\deg(r) < n$, and hence $\overline{g(X)} = \overline{r(X)}$. Thus, the set $\mathcal{B} = \{1, \overline{X}, \dots, \overline{X}^{n-1}\}$ is a generating set for V . It is also linearly independent, showing that $\mathcal{B}$ is a basis for V , and hence $\dim_k(V) = n$.

Theorem 1.2. Let k be a field, and $f(X) \in k[X]$ with $\deg f > 0$. There then exists a field $L \supset k$ such that $[L : k] := \dim_k L$ is finite, and $f(X)$ has a root in L .

Proof. Without the loss of generality, we may assume that $f(X)$ is irreducible over k . Write $f(X) = X^n + a_{n-1}X^{n-1} + \dots + a_0$. Consider the companion matrix of f :

$$C(f) = \begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{pmatrix} \quad (1.1)$$

This matrix may be obtained from the k -linear map $\ell_{\overline{X}} : k[X]/\langle f \rangle \rightarrow k[X]/\langle f \rangle$ defined by $\ell_{\overline{X}}(g(\overline{X})) = \overline{X}g(\overline{X})$; the matrix of $\ell_{\overline{X}}$ with respect to the basis $\mathcal{B}$ formed previously is precisely $C(f)$. Note that the minimal and characteristic polynomials of $C(f)$ are both equal to $f(X)$. Define $\alpha := C(f) \in M_n(k)$, and let $L := \{b_0 + b_1\alpha + \dots + b_{n-1}\alpha^{n-1} \mid b_i \in k\} \subseteq M_n(k)$. Then L is a ring. In fact, L is a field since any non-zero element's inverse can be computed using the division algorithm. Moreover, $f(\alpha) = 0$ and $L \supset k$.

As an alternate proof, setting $L := k[X]/\langle f \rangle$ gives a field extension of k since $\langle f \rangle$ is a maximal ideal. Moreover, $\overline{X} \in L$ is a root of $f(X)$. ■

1. Let R be a (commutative) ring, and $\{S_\lambda\}_{\lambda \in \Lambda}$ be an arbitrary collection of subrings of R . Then $\bigcap_{\lambda \in \Lambda} S_\lambda$ is also a subring of R .
2. In particular, the intersection of all subrings of R is a subring of R termed the *prime subring* of R denoted by $\mathfrak{p}_R$. It is also the image of the ring homomorphism $\mathbb{Z} \rightarrow R$ defined by $n \mapsto n \cdot 1_R$.
3. Let $R \subseteq S$ be rings, and $T \subseteq S$ be a subset. Define $R[T]$ as the intersection of all subrings of S containing R and T . It is precisely $\{f(t_1, \dots, t_r) \mid f \in R[X_1, \dots, X_r], r \geq 0, t_i \in T\}$.

If we let $k \subseteq L$ be fields, and $S \subseteq L$ be a subset, then $k[S]$ is an integral domain. We can then further define $k(S)$ as the field of fractions of $k[S]$, or

$$k(S) = \bigcap_{\substack{E \text{ field} \\ k \subseteq E \subseteq L \\ S \subseteq E}} E = \left\{ \frac{f(s_1, \dots, s_r)}{g(s_1, \dots, s_r)} \mid f, g \in k[X_1, \dots, X_r], g \neq 0, s_i \in S, r \geq 0 \right\}. \quad (1.2)$$

Thus, $k(S)$ is the smallest subfield of L containing k and S .

Definition 1.3. An element $\alpha \in \mathbb{C}$ is said to be an *algebraic number* (over $\mathbb{Q}$) if α is a root of a non-zero polynomial over $\mathbb{Q}$.

Definition 1.4. Define

$$\overline{\mathbb{Q}} := \{\alpha \in \mathbb{C} \mid \alpha \text{ is an algebraic number}\} \subseteq \mathbb{C}. \quad (1.3)$$

This is termed the *field of algebraic numbers*.

Of course, the fact that $\overline{\mathbb{Q}}$ is a field is not immediate. We prove minor results to support this claim.

Lemma 1.5. Let $\alpha \in \overline{\mathbb{Q}}$ and $m_\alpha(X)$ be the (a) non-zero minimal degree polynomial over $\mathbb{Q}$ with $m_\alpha(\alpha) = 0$. Then $m_\alpha(X)$ is irreducible over $\mathbb{Q}$.

Lemma 1.6. Let $\alpha \in \overline{\mathbb{Q}}$. Then $\mathbb{Q}[\alpha] = \mathbb{Q}(\alpha)$, and $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ is finite.

The concept of algebraic numbers can be generalized to any field k . Let $k \subseteq L$ be a field extension. An element $\alpha \in L$ is said to be algebraic over k if α is a root of a non-zero polynomial over k . Here, $k[\alpha] = k(\alpha)$, and $[k(\alpha) : k] = \deg m_{\alpha(X)}$ is finite.

Lemma 1.7. Let $E \supseteq L \supseteq k$ be fields. Then $[E : k] = [E : L][L : k]$.

Proof. If one of E/L or L/k is infinite dimensional, then E/k is infinite dimensional and the equality holds. Now assume E/k is finite dimensional. Clearly, $[E : k]$ being of finite dimension implies that $[E : L]$ and $[L : k]$ are finite dimensional. Let $\{e_1, \dots, e_m\}$ be an L -basis for E , and $\{\ell_1, \dots, \ell_n\}$ be a k -basis for L . We claim that $\{e_i \ell_j \mid 1 \leq i \leq m, 1 \leq j \leq n\}$ is a k -basis for E . It is clear that this set spans E over k . To show linear independence, assume $\sum_{i,j} a_{ij} e_i \ell_j = 0$ for some $a_{ij} \in k$. Then $\sum_i (\sum_j a_{ij} \ell_j) e_i = 0$. Since $\{e_1, \dots, e_m\}$ is linearly independent over L , we have $\sum_j a_{ij} \ell_j = 0$ for each i . Since $\{\ell_1, \dots, \ell_n\}$ is linearly independent over k , we have $a_{ij} = 0$ for all i, j . Thus, the set is linearly independent over k , and hence a basis. Therefore, $[E : k] = mn = [E : L][L : k]$. ■

Definition 1.8. A field extension L/k is called an *algebraic extension* if all elements of L are algebraic over k .

Theorem 1.9. Every finite extension of a field is algebraic.

Proof. Let L/k be a finite extension. Then for any $\alpha \in L$, set the left multiplication map $\ell_\alpha : L \rightarrow L$ defined by $\ell_\alpha(x) = \alpha x$. This is a k -linear map, and hence has a characteristic polynomial $P_\alpha(X)$. By the Cayley-Hamilton theorem, $P_\alpha(\ell_\alpha) = 0$, and hence $P_\alpha(\alpha) = 0$. Thus, α is algebraic over k . ■

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Theorem 1.10. *Let $E \supseteq L \supseteq k$ be fields. Then E/k is algebraic if and only if both E/L and L/k are algebraic.*

Such an inclusion is termed a *tower of extensions*. To prove this theorem, we require the following lemma.

Lemma 1.11. *Let $L \supseteq k$ be a field extension, and let $\alpha_1, \dots, \alpha_n \in L$ be algebraic over k . Then $k(\alpha_1, \dots, \alpha_n)$ is finite dimensional, and algebraic, over k .*

A field extension $L = k(\beta)$ is called a *simple extension*. In general, if L/k is a field extension, then L/k is called a *finitely generated extension* if there exists a finite subset $S \subseteq L$ such that $L = k(S)$. Note that a finitely generated extension need not be a finite extension. For example, $\mathbb{Q}(\pi)/\mathbb{Q}$ is finitely generated but not finite.

Proof of theorem. Let us assume L/E and E/k are algebraic; let $\alpha \in E$. Then α is algebraic over L , and there exists a non-zero polynomial $g(X) \in L[X]$ such that $g(\alpha) = 0$. Write $g(X) = a_0 + a_1X + \dots + a_rX^r$ with $a_i \in L$. Now consider $k(a_0, a_1, \dots, a_r)$; here, each a_i is algebraic over k so $k(a_0, a_1, \dots, a_r)/k$ is algebraic. Consider $k(a_0, a_1, \dots, a_r)(\alpha)$; since α is algebraic over $k(a_0, a_1, \dots, a_r)$, we have $[k(a_0, a_1, \dots, a_r)(\alpha) : k(a_0, a_1, \dots, a_r)] < \infty$. Thus, $[k(a_0, a_1, \dots, a_r)(\alpha) : k]$ is finite, and hence α is algebraic over k . Since $\alpha \in E$ was arbitrary, we have that E/k is algebraic. The converse is immediate since any $\alpha \in L$ is also in E , and hence algebraic over k . ■

Example 1.12. As an exercise, show that an algebraic extension L/k is finitely generated if and only if $[L : k]$ is finite.

1.2 Number Fields

Simply put, a finite field extension of the rationals $\mathbb{Q}$ is called a *number field*. Examples include $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\alpha_1, \dots, \alpha_r)$ where α_i are algebraic over $\mathbb{Q}$. Non-examples include $\mathbb{Q}(\pi)$, $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, \dots, \sqrt{p}, \dots)$ for all primes p . Recall the field of algebraic numbers:

$$\overline{\mathbb{Q}} = \{\alpha \in \mathbb{C} \mid \alpha \text{ is algebraic over } \mathbb{Q}\}. \quad (1.4)$$

It is left as an exercise to show that $\overline{\mathbb{Q}}$ is a field, and that $\overline{\mathbb{Q}}/\mathbb{Q}$ is an algebraic extension.

Definition 1.13. A field K is called *algebraically closed* if every non-constant polynomial $f(X) \in K[X]$ has a root in K . Equivalently, irreducible polynomials over K are exactly the linear polynomials.

For example, $\mathbb{C}$ is algebraically closed, but $\mathbb{R}$ is not. This is a consequence of the fundamental theorem of algebra.

Example 1.14. The field $\overline{\mathbb{Q}}$ is algebraically closed. To show this, let $f(X) \in \overline{\mathbb{Q}}[X]$ be an irreducible polynomial. By the fundamental theorem of algebra, there exists a root $\alpha \in \mathbb{C}$ of $f(X)$. So, $\overline{\mathbb{Q}}(\alpha)/\overline{\mathbb{Q}}$ is algebraic. But since $\overline{\mathbb{Q}}/\mathbb{Q}$ is algebraic, we have that $\overline{\mathbb{Q}}(\alpha)/\mathbb{Q}$ is algebraic. Thus, $\alpha \in \overline{\mathbb{Q}}$, and hence $f(X)$ has a root in $\overline{\mathbb{Q}}$.

If we define $\overline{\mathbb{Q}}^{\mathbb{R}} = \{\alpha \in \mathbb{R} \mid \alpha \text{ is algebraic over } \mathbb{Q}\}$. Clearly, $\overline{\mathbb{Q}}^{\mathbb{R}}$ is algebraic. Consider $i \in \mathbb{C}$; i is algebraic over $\mathbb{Q}$ but $i \notin \mathbb{R}$, so $i \notin \overline{\mathbb{Q}}^{\mathbb{R}}$. Thus, the minimal polynomial of i over $\overline{\mathbb{Q}}^{\mathbb{R}}$ is $X^2 + 1$, which is

irreducible over $\overline{\mathbb{Q}}^{\mathbb{R}}$. Hence, $\overline{\mathbb{Q}}^{\mathbb{R}}$ is not algebraically closed.

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Let $\varphi : F \rightarrow R$ be a ring homomorphism between a non-trivial field F and a non-trivial ring R . Since F is a field, it has only two ideals, $\{0\}$ and F . Enforcing that $\varphi(1) \neq 0$, we find that $\ker \varphi = \{0\}$, and hence φ is injective. Hence, on various occasions, we will identify F with its image $\varphi(F) \subseteq R$.

Theorem 1.15. *Let k be a field. Then there exists an algebraically closed field K containing k .*

To provide a sketch of a proof, pick, for each non-constant polynomial $f(X) \in k[X]$, a variable X_f . Consider the polynomial ring $R = k[X_f \mid f \in k[X], f \text{ non-constant}]$. Let I be the ideal generated by $\{f(X_f) \mid f \in k[X], f \text{ non-constant}\}$. This I is a proper ideal since it does not contain the unity. By Zorn's lemma, there exists a maximal ideal $\mathfrak{m} \supseteq I$. Now define $K = R/\mathfrak{m}$, a field. Since $k \hookrightarrow R \rightarrow K$, we have that K is a field extension of k . This K is *not* equal to k since the element $X_f + \mathfrak{m}$ is not in k but is algebraic over k since $f(X_f + \mathfrak{m}) = 0$ in K_0 . Hence, K/k is an algebraic extension. It now follows that every $f(X) \in k[X]$ with degree at least 1 has a root in $K = K_0$. Repeating this process, we can construct a tower of algebraic extensions $K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots$, and define $E = \bigcup_{n=0}^{\infty} K_n$. Then E/k is algebraic, and every non-constant polynomial in $E[X]$ has a root in E . Hence, E is algebraically closed.

Definition 1.16. Let k be a field. Then the *algebraic closure* of k is an algebraically closed field $\bar{k}$ containing k such that $\bar{k}/k$ is algebraic.

We know that if $L \supseteq k$ is a field extension, then L is also a k -vector space. For any algebraic closure $\bar{k}$ of k (or an algebraically closed field containing k), a homomorphism $\varphi : L \rightarrow \bar{k}$ is called a *k -embedding* if φ is k -linear homomorphism of fields.

Example 1.17. We look at some examples of k -embeddings.

- Let $k = \mathbb{R}$ and $\bar{k} = \mathbb{C}$. With $L = \mathbb{R}(\sqrt{-2})$, which is essentially $\mathbb{C}$, calling $\alpha = \sqrt{-2}$, we have that $\varphi(a + b\alpha) = a + b\bar{\alpha}$ and $\varphi(a + b\alpha) = a + b\alpha$ are the only two $\mathbb{R}$ -embeddings of L into $\mathbb{C}$.
- Let α be algebraic over k , and let $E = k(\alpha)$. We ask what are all the k -embeddings of E into $\bar{k}$. Let $m_{\alpha}(X)$ be the minimal polynomial of α over k . Pick any root $\beta \in \bar{k}$ of $m_{\alpha}(X)$, and let $\deg m_{\alpha}(X) = d$. Define a mapping $\varphi : k(\alpha) \rightarrow \bar{k}$ by $\alpha \mapsto \beta$, and extending k -linearly. Then φ is a k -embedding of E into $\bar{k}$. Conversely, if $\sigma : k(\alpha) \rightarrow \bar{k}$ is a k -embedding, then $\sigma(\alpha)$ is a root of $m_{\alpha}(X)$ in $\bar{k}$. Thus, the k -embeddings of E into $\bar{k}$ are in bijection with the distinct roots of $m_{\alpha}(X)$ in $\bar{k}$.

July 27th.

Definition 1.18. Let $L \supseteq k$ be a field extension. An element $\alpha \in L$ is said to be *separable* over k if the minimal polynomial of α over k has roots distinct in $\bar{k}$. The field extension L/k is said to be *separable* if every element of L is separable over k .

Let $f \in k[X]$ be irreducible. Then f has distinct roots in some algebraic closure $\bar{k}$ of k if and only if $\text{char}(k) = 0$, or $\text{char}(k) = p > 0$ and $f(X) = g(X^p)$ for some $g \in k[X]$.

For $f(X) = a_0 + a_1X + a_2X^2 + \cdots + a_rX^r$, we define the *formal derivative* of f as

$$f'(X) := a_1 + 2a_2X + 3a_3X^2 + \cdots + ra_rX^{r-1}. \quad (1.5)$$

Check that $(fg)' = f'g + fg'$ and $(f + g)' = f' + g'$.

Example 1.19. We look at a couple examples.

1. Any number field $K \supseteq \mathbb{Q}$ is separable since $\text{char}(\mathbb{Q}) = 0$.
2. Let $\mathbb{F}_q$ be the finite field with $q = p^n$ elements. Then any algebraic extension of $\mathbb{F}_q$ is separable. In fact, a field k is called *perfect* if all algebraic extensions of k are separable.

3. As a non-example, let $k = \mathbb{F}_p(X)$ and $K = k(X^{1/p})$. Then K/k is not separable since $Y^p - X$, the polynomial in $k[Y]$, has a double root in $\bar{k}$. Such an extension is called *inseparable*.

Theorem 1.20. *Let L be a finite extension of k . Assume that L is separable over k . Then there exists an $\alpha \in L$ such that $L = k(\alpha)$. In other words, any finite separable extension is a simple extension.*

Theorem 1.21. *Let k be a field, and K_1 and K_2 be two algebraic closures of k . Then there exists a k -isomorphism from K_1 to K_2 . In particular, the algebraic closure of a field is unique up to k -isomorphism.*

Proof. Let k be a field, and L an algebraically closed field. Let $E \supseteq k$ be an algebraic extension. Then any embedding $\sigma : k \rightarrow L$ extends to an embedding $\tilde{\sigma} : E \rightarrow L$; we shall utilize this result. If $i : k \rightarrow K_2$ is the inclusion map, then there exists an embedding $\tilde{i} : K_1 \rightarrow K_2$ extending i . This $\tilde{i}$ also happens to be surjective since $K_2/\tilde{i}(K_1)$ and $\tilde{i}(K_1)/k$ are algebraic extensions, and $\tilde{i}(K_1)$ is algebraically closed. Hence, $\tilde{i}$ is a k -isomorphism from K_1 to K_2 . ■

The result used in the proof above is a consequence of Zorn's lemma. Let k be a field, L an algebraically closed field, and $E \supseteq k$ an algebraic extension. Moreover, let $\sigma : k \rightarrow L$ be an embedding. Consider

$$\mathcal{F} = \{(M, \sigma_M) \mid E \supseteq M \supseteq k, \sigma_M : M \rightarrow L \text{ is an embedding extending } \sigma\}. \quad (1.6)$$

Define a partial order on here as $(M, \sigma_M) \succeq (N, \sigma_N)$ if $M \supseteq N$ and $\sigma_M|_N = \sigma_N$. Let $\{(M_\lambda, \sigma_\lambda)\}_{\lambda \in \Lambda}$ be a chain in $\mathcal{F}$. Then we can define $M = \bigcup_{\lambda \in \Lambda} M_\lambda$, and $\sigma : M \rightarrow L$ by $\sigma|_{M_\lambda} = \sigma_\lambda$. Then $(M, \sigma) \in \mathcal{F}$, and $(M, \sigma) \succeq (M_\lambda, \sigma_\lambda)$ for all λ . By Zorn's lemma, there exists a maximal element (M, σ_M) in $\mathcal{F}$. We claim that $M = E$. If not, pick $\alpha \in E \setminus M$. Then $m_\alpha(X)$ is the minimal polynomial of α over M , and since L is algebraically closed, there exists a root $\beta \in L$ of $m_\alpha(X)$. Define an extension $\tilde{\sigma} : M(\alpha) \rightarrow L$ by $\tilde{\sigma}|_M = \sigma_M$ and $\tilde{\sigma}(\alpha) = \beta$. This contradicts the maximality of (M, σ_M) , and hence $M = E$. Thus, any embedding of k into an algebraically closed field extends to an embedding of any algebraic extension of k .

For a field k , let $\alpha, \beta \in \bar{k}$. We wish to procure a necessary and sufficient condition for $k(\alpha) \cong k(\beta)$ as extensions of k , via an isomorphism φ with $\alpha \mapsto \beta$.

Lemma 1.22. *There exists a k -isomorphism $\varphi : k(\alpha) \rightarrow k(\beta)$ with $\varphi(\alpha) = \beta$ if and only if $m_\alpha(X) = m_\beta(X)$.*

Definition 1.23. Let k be a field, and fix an algebraic closure $\bar{k}$ of k . Let $f(X) \in k[X]$ be a polynomial. The *splitting field* of $f(X)$ over k is the smallest field extension of k containing all the roots of $f(X)$; that is,

$$\text{split}(f(X); k) = k(\alpha_1, \dots, \alpha_r) \quad (1.7)$$

where α_i are all roots of f in $\bar{k}$.

Let $f(X), k, \bar{k}$ be as above, and let $g(X) \in k[X]$ be an irreducible polynomial and suppose $g(X)$ has a root in $L = \text{split}(f(X); k)$. Then $g(X)$ splits completely in L .

Proof. Let β be a root of g in $\bar{k}$. Then there exists a k -isomorphism $\varphi : k(\alpha) \rightarrow k(\beta)$ with $\varphi(\alpha) = \beta$, where α is such that $L \supseteq k(\alpha) \supseteq k$. By the embedding extension property, φ extends to a k -embedding $\tilde{\varphi} : L \rightarrow \bar{k}$. Since L contains all roots of $f(X)$, we have that $\tilde{\varphi}(L)$ contains all roots of $f(X)$. But $\tilde{\varphi}(L) = L$, and hence L contains all roots of $g(X)$. ■

One can also talk about the splitting field of a set of polynomials too.

Definition 1.24. Let S be a set of polynomials in $k[X]$. Then, we simply define $\text{split}(S; k)$ to be the smallest field extension of k containing all the roots of $f(X)$, where $f(X) \in S$.

Here is another property of splitting fields; let $L = \text{split}(S; k) \hookrightarrow \bar{k}$. Then any k -embedding $\varphi : L \rightarrow \bar{k}$. Then $\varphi(L) = L$.

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